Detecting Lies in Video

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Chunk based processing

To extend PECL for long-form video deception detection, we introduce a Hierarchical Temporal Processing framework. First, Whisper segments videos into 2-5 second clips, ensuring alignment with speech timestamps. Each chunk is processed independently using PECL's audio-visual encoders (Wav2Vec2 + ViT), fused via PAVF, and classified at the chunk level. To model long-range dependencies, we employ a sequence-level processing stage, utilizing either (i) a Transformer with cross-attention for capturing global context, or (ii) a sliding-window Transformer with a memory bank for adaptive temporal focus. This enables deception classification across entire videos while providing attention-based interpretability to highlight deceptive segments.

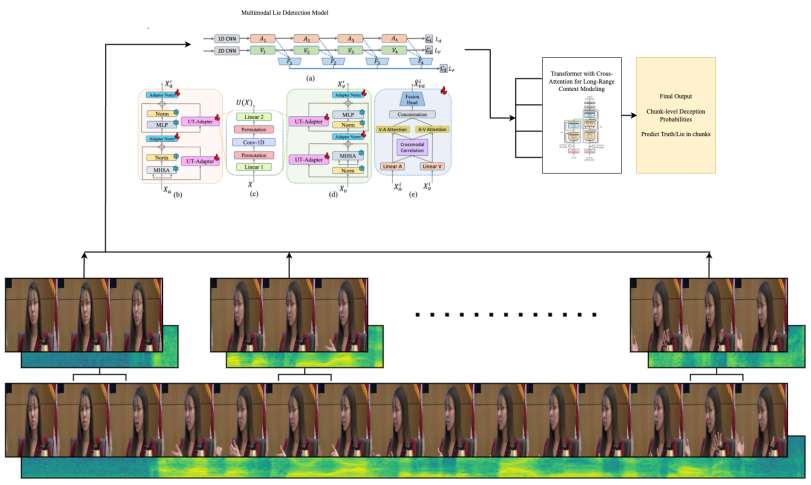


Figure 1. Chunk Processing for long videos.

Video Processing

Transformer-based models like TimeSformer [2] and Video Swin Transformer [3] apply self-attention to learn long-range dependencies. Hybrid approaches, such as ViViT[1] and X-CLIP [4], integrate transformers with CNN backbones for efficient video understanding.

Why chunk videos to smaller clips?

Not only does it make the task for the model much simpler (binary classification vs multi-label), there's also:

- **Long-range dependencies:** Capturing temporal coherence in extended videos is challenging.
- **Model limitations:** RNNs suffer from vanishing gradients, while transformers face quadratic complexity.
- **Multimodal fusion:** Aligning video, audio, and text over long durations increases complexity.
- **Efficient processing:** Chunking, memory mechanisms, and hierarchical models help mitigate these issues.

Baseline Results

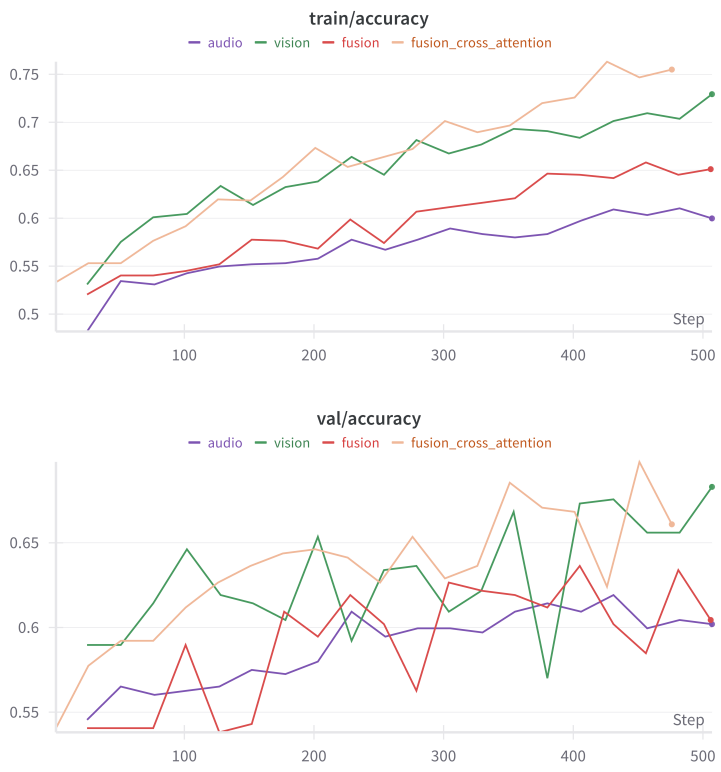


Figure 2. Train and Validation Accuracy.

Qualitative Results

Figure 3 shows the activation map of the first CNN layer, highlighting the regions of interest detected by the model.

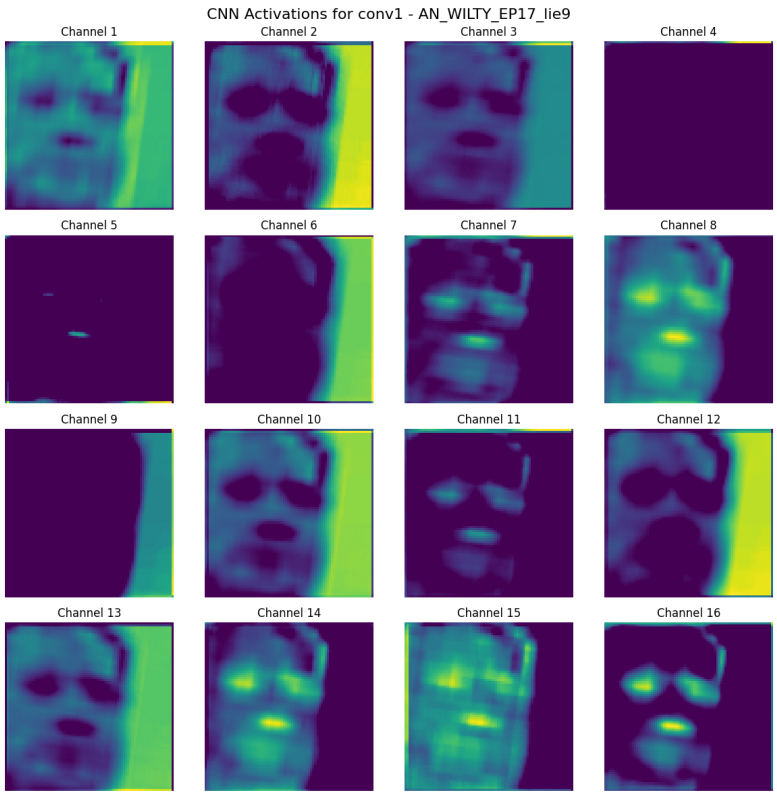


Figure 3. CNN layer-1 activation map.

Multi-task Learning

Multi-task Learning enhances model performance by simultaneously learning multiple tasks, allowing shared representations to improve generalization. In this framework, we predict K + 1 binary labels, where K labels correspond to MUMIN features and last is for lie/truth prediction. The fused multimodal embedding is used for classification. The multi-task loss function is formulated as follows:

$$L_M = -\sum_{k=1}^{K+1} \left(Y_k \log(S_k) + (1 - Y_k) \log(1 - S_k)\right)$$

Interesting Cases

Interestingly, when we trained XGBoost using only these MUMIN features, we achieved 65% accuracy, indicating that these features alone can capture enough information to get results as good as the base fusion model. This suggests that the MUMIN features may contain significant latent information, and if a deep learning model can predict these features more effectively, it could serve as an inherent feature selector.

Future work

3D Vision Transformer for Spatiotemporal Analysis We plan to explore a **3D Vision Transformer (ViT)** to better capture temporal dependencies in deception detection. This approach would tokenize video frames into spatiotemporal patches, leveraging self-attention mechanisms to model motion cues and expression dynamics.

Enhancing the Classifier with Downstream Tasks The DOLOS dataset contains auxiliary labels (e.g., facial expressions). We aim to integrate **multi-task learning (MTL)** by predicting deception alongside these labels, using an **MLP classifier** trained on learned embeddings to refine predictions and improve model robustness.

Framework We dedicated significant time to refactoring the existing code, addressing compatibility issues with libraries, deprecated dependencies, and correcting architectural mistakes, such as missing activation functions. We aim to further enhance the framework and integrate a graphical interface for **real-time video processing**.

References

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[3] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9992–10002, 2021.

[4] Bolin Ni, Houwen Peng, Minghao Chen, Songyang Zhang, Gaofeng Meng, Jianlong Fu, Shiming Xiang, and Haibin Ling. Expanding language-image pretrained models for general video recognition, 2022.